

SHADRACK MUTETHIA

Full-Stack Software Engineer

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[LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Full-Stack Software Engineer with 3+ years of experience building scalable web applications using MERN/PERN stacks and modern JavaScript frameworks, currently building FHIR-native healthcare systems at Tiberbu Healthnet Limited. Expertise in developing RESTful APIs, optimizing database performance, and deploying cloud-based solutions on AWS and Azure. Proven track record of delivering high-performance applications with 99.9% uptime and reducing API response times by up to 45%. Strong foundation in data structures, algorithms, and system design. Experienced in healthcare interoperability standards, asynchronous communication, and remote collaboration with distributed teams.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C, C++, SQL, HTML5, CSS3

Frontend: React.js, Next.js, Vite, Tailwind CSS, Bootstrap, Redux, Tanstack, Zustand, React Router

Backend: Node.js, Express.js, Frappe Framework, RESTful APIs, JWT Authentication, Microservices

Databases: PostgreSQL, MongoDB, MySQL, MariaDB, Redis, Database Optimization, Query Performance

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda), Azure, Docker, Kubernetes, CI/CD, Jenkins, GitHub Actions, Kargo

Monitoring: Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana)

Tools: Git, GitHub, Postman, Bruno, VS Code, Web-storm, BMAD

Testing: Jest, React Testing Library, Code Reviews, Performance Testing

RELEVANT PROFESSIONAL EXPERIENCE

Full-Stack Developer (Contract)

Tiberbu Healthnet | Nairobi, Kenya | January 2026 – Present

- Build full-stack healthcare management systems using the Frappe framework for backend development and Next.js for frontend applications
- Design and develop FHIR-compliant APIs and resources (e.g., Patient, Observation, Encounter) to enable interoperability between healthcare systems
- Architect FHIR-native data models and workflows that support standards-based health information exchange
- Collaborate with clinical stakeholders and cross-functional teams to translate healthcare workflows into scalable, compliant software solutions
- Integrate Frappe's Python-based backend with modern Next.js frontend architecture to deliver responsive, performant healthcare applications

Full-Stack Developer (Remote Freelance)

Self-Employed | Nairobi, Kenya | February 2023 – Present

- Develop full-stack web applications remotely for international clients using MERN and PERN stacks, achieving 99.9% uptime and sub-2-second page load times
- Build RESTful APIs and microservices with Node.js and Express.js, implementing JWT authentication and role-based access control
- Design and optimize PostgreSQL and MongoDB databases with indexing strategies, improving query performance by 40%
- Deploy containerized applications using Docker and Kubernetes on AWS (EC2, S3, RDS) and Azure with CI/CD pipelines
- Implement Redis caching to reduce database load by 35% and improve API response times by 45%
- Configure monitoring solutions using Prometheus, Grafana, and ELK Stack for real-time performance tracking
- Develop React applications with Next.js and Vite, utilizing server-side rendering (SSR) and code splitting for optimal performance
- Collaborate with cross-functional teams using Agile methodologies, Git workflows, and asynchronous communication tools (Slack, Jira)

Key Projects: E-commerce platform with Stripe integration and admin dashboard | Real-time analytics dashboard with PostgreSQL and WebSocket | Multi-tenant SaaS application with subscription management

Technical Support Engineer

XENO Technologies Ltd | Nairobi, Kenya | June 2021 – October 2023

- Provided technical support for web applications and APIs, troubleshooting frontend and backend issues to resolve 95% of tickets within SLA
- Developed technical documentation, user manuals, and API integration guides, reducing support tickets by 40%
- Analyzed application logs and performance metrics to identify bottlenecks and recommend optimization strategies
- Collaborated with development teams using bug tracking systems (Jira) to escalate and resolve complex technical issues
- Assisted customers with database queries, API authentication issues, and deployment configurations
- Created automated scripts in Python and JavaScript to streamline common support tasks and improve response times

Energy Engineer | Auditor

SYRECON SERVICES LIMITED | Nairobi, Kenya | Jan 2022 – June 2023

- Evaluated electrical systems, HVAC units, and other power equipment to identify energy savings opportunities that enhanced system performance and reduced costs.
- Developed comprehensive audit reports, including technical specifications and cost-benefit analyses, enabling clients to make data-driven decisions on energy-efficient retrofits and switchgear upgrades.
- Consulted with clients on transformers, distribution boards (DBs), and other equipment to optimize power distribution, ensuring compliance with industry standards.
- Collaborated with sales, engineering, and operations teams to structure customer proposals and secure buy-in for recommended solutions.

Energy Engineer

EIT AFRICA LIMITED | Nairobi, Kenya | May 2019 – Aug 2020

- Evaluated electrical systems, HVAC units, and other power equipment to identify energy savings opportunities that enhanced system performance and reduced costs.
 - Prepared bid proposals for presentation to potential clients under supervision of senior engineer
 - Assisted senior engineers in designing and simulating solar photovoltaic and hybrid power systems, building foundational knowledge of power transmission, distribution, and industrial controls.
 - Interpreted technical drawings and schematics related to switchgear, circuit breakers, and motor control systems to inform project scope and bid proposals.
 - Gained exposure to LV/HV electrical projects, providing estimates and cost analyses that aligned with project objectives and budgets.
 - Participated in client presentations to discuss system capabilities, hardware requirements, and expected performance metrics, fostering early sales success.
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EDUCATION

Bachelor of Science in Energy Technology

Kenyatta University, School of Engineering | Nairobi, Kenya | 2016 – 2022

Relevant Coursework: Programming (C, C++, Python), Data Structures & Algorithms, Database Systems, Software Development, Systems Design

The Odin Project - Full Stack JavaScript Path

Self-Paced Online Curriculum | 2022 – 2023

Comprehensive full-stack web development program covering HTML, CSS, JavaScript, React, Node.js, Express, MongoDB, PostgreSQL, Git, testing, and deployment. Completed hands-on projects including responsive websites, RESTful APIs, and full-stack applications.

CERTIFICATIONS

- Software Engineer | HackerRank | October 2025
- JavaScript Mastery | HackerRank | October 2025
- Artificial Intelligence Analyst | IBM | September 2019

FEATURED PROJECTS

AI-Powered ATS Resume Checker

Tech Stack: React, TypeScript, Node.js, Express.js, Tailwind CSS, Puter.js

GitHub: github.com/Slimdaddy254/resume-checker **Live:** resume-cheq.vercel.app/

- Built full-stack resume analysis tool with AI-powered ATS compatibility scoring and keyword optimization
- Developed Express.js RESTful API for file processing and analysis with error handling
- Implemented TypeScript across frontend and backend for type safety and improved maintainability
- Deployed with CI/CD pipeline on Vercel with automated testing

CV Builder Application

Tech Stack: React, TypeScript, Node.js, Express.js, MongoDB, CSS3

GitHub: github.com/Slimdaddy254/cv-builder | **Live:** cv-builder-rose.vercel.app

- Developed MERN stack resume builder with MongoDB for user data persistence and template management
- Built Express.js API for CRUD operations, JWT authentication, and PDF generation
- Created dynamic forms with real-time preview and auto-save functionality
- Designed print-optimized layouts for professional PDF export

File Storage Application (Google Drive Clone)

Tech Stack: Node.js, Express.js, Prisma, PostgreSQL, Cloudinary, JWT

GitHub: github.com/Slimdaddy254/file-uploader | **Live:** file-uploader-3gqw.onrender.com

- Built full-featured file storage system with folder management, file uploads, and shareable links similar to Google Drive
- Developed RESTful API using Express.js and Prisma ORM for database operations and file management
- Implemented PostgreSQL database schema with relationships for users, folders, files, and sharing permissions
- Integrated Cloudinary for cloud-based file storage and media management with automated backups
- Built JWT authentication system with user registration, login, and session management

- Created shareable link functionality with configurable expiration dates and access controls
 - Optimized file upload performance with multipart form handling and progress tracking
 - Implemented folder hierarchy system allowing nested folder creation and file organization
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ADDITIONAL INFORMATION

Remote Work: 3+ years of remote freelance experience with international clients across multiple time zones

Problem Solving: Strong analytical skills with expertise in data structures, algorithms, and system design

Performance Optimization: Proven track record of reducing latency and improving application responsiveness

Cloud-Native Development: Experience building secure, scalable solutions on AWS and Azure

Communication: Excellent written and verbal communication skills for remote collaboration and technical documentation

Languages: English (Fluent), Swahili (Native)